11.2 Cache deployment options

Which deployment option you should choose is influenced by the following four factors:

- *Engine*—Which in-memory database do you want to use: Memcached or Redis?
- *Backup/Restore*—Does your workload require data persistence, which means being able to backup and restore data?
- *Replication*—Is high availability important to your workload? If so, you need to replicate to at least one other node.
- *Sharding*—Does your data exceed the maximum memory available for a single node, or do you need to increase the throughput of your system?

Table 11.2 compares the deployment options for Memcached and Redis.

Next, we'll look at deployment options in more detail.

Table 11.2 Comparing ElastiCache and MemoryDB engines and deployment options

Service	Engine	Deployment Option	Backup/Restore	Replication	Sharding
ElastiCache	Memcached	Default	No	No	Yes
	Redis	Single Node	Yes	No	No
		Cluster Mode **disabled**	Yes	Yes	No
		Cluster Mode **enabled**	Yes	Yes	Yes
MemoryDB	Redis	Default	Yes	Yes	Yes

11.2.1 Memcached: Cluster

An Amazon ElastiCache for Memcached cluster consists of 1–40 nodes. Sharding is implemented by the Memcached client, typically using a consistent hashing algorithm, which arranges keys into partitions in a ring distributed across the nodes. The client decides which keys belong to which nodes and directs the requests to